

Environment

```
import torch
import torch.nn as nn
import torch.nn.functional as F
import torchvision
from torch.utils.data import DataLoader
from torchvision import datasets, transforms
import matplotlib.pyplot as plt
import numpy as np
from sklearn.decomposition import PCA
from scipy.stats import wasserstein_distance, ks_2samp
from scipy import stats
from tqdm import tqdm

torch.manual_seed(42)
np.random.seed(42)

device = 'cuda' if torch.cuda.is_available() else 'cpu'
print(f"Using device: {device}")

def plot_results(images, nrow=10, title=None):
    images = (images.cpu().detach().clone() * 0.5 + 0.5).clamp(0, 1)
    grid = torchvision.utils.make_grid(images, nrow=nrow, padding=1,
    normalize=False)
    plt.figure(figsize=(10, (images.size(0) // nrow + 1) * 1.0),
    dpi=300)
    np_grid = grid.permute(1, 2, 0).numpy()
    plt.imshow(np_grid)
    plt.axis("off")
    if title is not None:
        plt.title(title, fontsize=16)
    plt.show()

Using device: cuda
```

VAE Model

```
class PreActResidualBlock(nn.Module):
    """Pre-activation ResBlock: GN -> SiLU -> Conv -> GN -> SiLU ->
    Conv + shortcut."""
    def __init__(self, in_channels, out_channels, groups=8,
    dropout=0.1):
        super().__init__()
        self.gn1 = nn.GroupNorm(groups, in_channels)
        self.act1 = nn.SiLU()
        self.conv1 = nn.Conv2d(in_channels, out_channels, 3,
    padding=1, bias=False)
```

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        self.gn2 = nn.GroupNorm(groups, out_channels)
        self.act2 = nn.SiLU()
        self.dropout = nn.Dropout2d(dropout) if dropout > 0 else
nn.Identity()
        self.conv2 = nn.Conv2d(out_channels, out_channels, 3,
padding=1, bias=False)

        self.shortcut = (
            nn.Conv2d(in_channels, out_channels, 1, bias=False)
            if in_channels != out_channels else nn.Identity()
        )

    def forward(self, x):
        out = self.conv1(self.act1(self.gn1(x)))
        out = self.conv2(self.dropout(self.act2(self.gn2(out))))
        return out + self.shortcut(x)

class ConvVAE(nn.Module):
    def __init__(self, latent_dim=128, img_channels=1, groups=8,
dropout=0.1,
                    out_activation="tanh"): # out_activation: "tanh" or
"sigmoid"
        super().__init__()
        self.latent_dim = latent_dim
        self.img_channels = img_channels
        self.out_activation = out_activation

        # -----
        # Encoder: 28 -> 14 -> 7 -> 4
        # -----
        self.enc = nn.Sequential(
            # 28 -> 14
            nn.Conv2d(img_channels, 32, 3, stride=2, padding=1,
bias=False),
            PreActResidualBlock(32, 32, groups=groups,
dropout=dropout),

            # 14 -> 7
            nn.Conv2d(32, 64, 3, stride=2, padding=1, bias=False),
            PreActResidualBlock(64, 64, groups=groups,
dropout=dropout),

            # 7 -> 4
            nn.Conv2d(64, 128, 3, stride=2, padding=1, bias=False),
            PreActResidualBlock(128, 128, groups=groups,
dropout=dropout),
        )
        self.mid = PreActResidualBlock(128, 128, groups=groups,
dropout=dropout)

```

```

self.flatten_dim = 128 * 4 * 4
self.fc_mu = nn.Linear(self.flatten_dim, latent_dim)
self.fc_logvar = nn.Linear(self.flatten_dim, latent_dim)

# -----
# Decoder: z -> 4x4 -> 7 -> 14 -> 28
# -----
self.fc_decode = nn.Linear(latent_dim, self.flatten_dim)

self.dec1 = nn.Sequential(
    # 4 -> 7 (k=3,s=2,p=1 gives 7)
    nn.ConvTranspose2d(128, 64, 3, stride=2, padding=1,
output_padding=0, bias=False),
    nn.GroupNorm(groups, 64),
    nn.SiLU(),
    PreActResidualBlock(64, 64, groups=groups,
dropout=dropout),
)
self.dec2 = nn.Sequential(
    # 7 -> 14
    nn.ConvTranspose2d(64, 32, 4, stride=2, padding=1,
bias=False),
    nn.GroupNorm(groups, 32),
    nn.SiLU(),
    PreActResidualBlock(32, 32, groups=groups,
dropout=dropout),
)
self.dec3 = nn.Sequential(
    # 14 -> 28
    nn.ConvTranspose2d(32, 32, 4, stride=2, padding=1,
bias=False),
    nn.GroupNorm(groups, 32),
    nn.SiLU(),
)
self.out_conv = nn.Conv2d(32, img_channels, 3, padding=1)

self.apply(self._init_weights)

@staticmethod
def _init_weights(m):
    if isinstance(m, (nn.Conv2d, nn.ConvTranspose2d)):
        nn.init.kaiming_normal_(m.weight, nonlinearity='relu')
        if getattr(m, 'bias', None) is not None:
            nn.init.zeros_(m.bias)
    if isinstance(m, nn.Linear):
        nn.init.xavier_uniform_(m.weight)
        if m.bias is not None:
            nn.init.zeros_(m.bias)

```

```

def encode(self, x):
    h = self.enc(x)
    h = self.mid(h)
    h = h.view(h.size(0), -1)
    mu = self.fc_mu(h)
    logvar = self.fc_logvar(h)
    return mu, logvar

def reparameterize(self, mu, logvar):
    std = torch.exp(0.5 * logvar)
    eps = torch.randn_like(std)
    return mu + eps * std

def decode(self, z):
    h = self.fc_decode(z)
    h = h.view(h.size(0), 128, 4, 4)
    h = self.dec1(h)
    h = self.dec2(h)
    h = self.dec3(h)
    h = self.out_conv(h)
    if self.out_activation == "sigmoid":
        return torch.sigmoid(h)
    else:
        return torch.tanh(h)

def forward(self, x):
    mu, logvar = self.encode(x)
    z = self.reparameterize(mu, logvar)
    recon = self.decode(z)
    return recon, mu, logvar

```

Loss Function & Training Script

```

def vae_loss_function(recon_x, x, mu, logvar, beta=1.0):
    recon_loss = F.mse_loss(recon_x, x, reduction='sum')
    kld = -0.5 * torch.sum(1 + logvar - mu.pow(2) - logvar.exp())
    return recon_loss + beta * kld, recon_loss, kld

def train_conv_vae(vae, train_loader, epochs=20, device='cuda',
beta=1.0, lr=1e-3):
    vae = vae.to(device)
    optimizer = torch.optim.AdamW(vae.parameters(), lr=lr,
weight_decay=1e-5)
    scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(optimizer,
T_max=epochs)

    print("Training Convolutional VAE...")
    print(f"Model parameters: {sum(p.numel() for p in
vae.parameters()):,}")

```

```

history = {"loss": [], "recon": [], "kld": []}

for epoch in range(epochs):
    vae.train()
    train_loss = 0
    train_recon = 0
    train_kld = 0

    pbar = tqdm(train_loader, desc=f"Epoch {epoch+1}/{epochs}")
    for batch_idx, (data, _) in enumerate(pbar):
        data = data.to(device)
        optimizer.zero_grad()

        recon_batch, mu, logvar = vae(data)
        loss, recon_loss, kld = vae_loss_function(recon_batch,
data, mu, logvar, beta)

        loss.backward()
        torch.nn.utils.clip_grad_norm_(vae.parameters(),
max_norm=1.0)
        optimizer.step()

        train_loss += loss.item()
        train_recon += recon_loss.item()
        train_kld += kld.item()

    pbar.set_postfix({
        'loss': f'{loss.item()/len(data):.2f}',
        'recon': f'{recon_loss.item()/len(data):.2f}',
        'kld': f'{kld.item()/len(data):.2f}'
    })

    n_samples = len(train_loader.dataset)
    avg_loss = train_loss / n_samples
    avg_recon = train_recon / n_samples
    avg_kld = train_kld / n_samples

    scheduler.step()
    current_lr = optimizer.param_groups[0]['lr']

    history['loss'].append(avg_loss)
    history['recon'].append(avg_recon)
    history['kld'].append(avg_kld)

    print(f'Epoch {epoch+1}/{epochs}:')
    print(f'  Total Loss: {avg_loss:.4f}')
    print(f'  Recon Loss: {avg_recon:.4f}')
    print(f'  KLD Loss: {avg_kld:.4f}')
    print(f'  LR: {current_lr:.6f}')

```

```
return vae, history
```

Load MNIST Dataset

```
print("="*60)
print("STEP 1: Loading MNIST Dataset")
print("="*60)

transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.5,), (0.5,))
])

train_dataset = datasets.MNIST('./data', train=True, download=True,
                                transform=transform)
train_loader = DataLoader(train_dataset, batch_size=128, shuffle=True,
                           num_workers=4, pin_memory=True)

print(f"Dataset loaded: {len(train_dataset)} training images")

=====
STEP 1: Loading MNIST Dataset
=====

100%|██████████| 9.91M/9.91M [00:01<00:00, 5.34MB/s]
100%|██████████| 28.9k/28.9k [00:00<00:00, 131kB/s]
100%|██████████| 1.65M/1.65M [00:01<00:00, 1.25MB/s]
100%|██████████| 4.54k/4.54k [00:00<00:00, 9.21MB/s]

Dataset loaded: 60000 training images

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py
:627: UserWarning: This DataLoader will create 4 worker processes in
total. Our suggested max number of worker in current system is 2,
which is smaller than what this DataLoader is going to create. Please
be aware that excessive worker creation might get DataLoader running
slow or even freeze, lower the worker number to avoid potential
slowness/freeze if necessary.
  warnings.warn(
```

Train and Save the Model

```
print("\n" + "="*60)
print("STEP 2: Training Convolutional VAE")
print("="*60)

latent_dim = 128
vae = ConvVAE(latent_dim=latent_dim, img_channels=1)
```

```
beta = 3
vae, history = train_conv_vae(vae, train_loader, epochs=40,
device=device, beta=beta, lr=1e-3)

torch.save({
    'model_state_dict': vae.state_dict(),
    'latent_dim': latent_dim,
    'history': history,
}, f'conv_vae_checkpoint_{beta}.pt')
print("\n ✓ VAE checkpoint saved!")
```

=====

STEP 2: Training Convolutional VAE

=====

Training Convolutional VAE...
Model parameters: 1,780,545

Epoch 1/40: 100%|██████████| 469/469 [00:19<00:00, 24.02it/s,
loss=140.09, recon=92.65, kld=15.81]

Epoch 1/40:
Total Loss: 1236.7399
Recon Loss: 126.2827
KLD Loss: 370.1524
LR: 0.000998

Epoch 2/40: 100%|██████████| 469/469 [00:18<00:00, 25.54it/s,
loss=119.22, recon=77.34, kld=13.96]

Epoch 2/40:
Total Loss: 122.1841
Recon Loss: 77.8739
KLD Loss: 14.7700
LR: 0.000994

Epoch 3/40: 100%|██████████| 469/469 [00:19<00:00, 24.57it/s,
loss=103.66, recon=57.47, kld=15.40]

Epoch 3/40:
Total Loss: 114.6437
Recon Loss: 70.4328
KLD Loss: 14.7369
LR: 0.000986

Epoch 4/40: 100%|██████████| 469/469 [00:18<00:00, 25.58it/s,
loss=100.63, recon=57.98, kld=14.22]

Epoch 4/40:
Total Loss: 110.2501
Recon Loss: 66.0580

KLD Loss: 14.7307
LR: 0.000976

Epoch 5/40: 100%|██████████| 469/469 [00:20<00:00, 23.23it/s,
loss=110.03, recon=62.89, kld=15.71]

Epoch 5/40:
Total Loss: 107.6646
Recon Loss: 63.1924
KLD Loss: 14.8241
LR: 0.000962

Epoch 6/40: 100%|██████████| 469/469 [00:18<00:00, 25.84it/s,
loss=102.72, recon=58.93, kld=14.60]

Epoch 6/40:
Total Loss: 105.6845
Recon Loss: 61.1686
KLD Loss: 14.8386
LR: 0.000946

Epoch 7/40: 100%|██████████| 469/469 [00:18<00:00, 25.41it/s,
loss=102.87, recon=59.41, kld=14.49]

Epoch 7/40:
Total Loss: 104.0518
Recon Loss: 59.6089
KLD Loss: 14.8143
LR: 0.000926

Epoch 8/40: 100%|██████████| 469/469 [00:19<00:00, 24.36it/s,
loss=103.22, recon=58.12, kld=15.04]

Epoch 8/40:
Total Loss: 102.8743
Recon Loss: 58.2124
KLD Loss: 14.8873
LR: 0.000905

Epoch 9/40: 100%|██████████| 469/469 [00:18<00:00, 25.94it/s,
loss=98.75, recon=54.81, kld=14.65]

Epoch 9/40:
Total Loss: 101.7749
Recon Loss: 57.0792
KLD Loss: 14.8986
LR: 0.000880

Epoch 10/40: 100%|██████████| 469/469 [00:19<00:00, 24.54it/s,
loss=101.56, recon=56.78, kld=14.93]

Epoch 10/40:

Total Loss: 100.8372
Recon Loss: 56.1970
KLD Loss: 14.8801
LR: 0.000854

Epoch 11/40: 100%|██████████| 469/469 [00:18<00:00, 25.49it/s,
loss=103.36, recon=58.23, kld=15.05]

Epoch 11/40:

Total Loss: 100.0669
Recon Loss: 55.3520
KLD Loss: 14.9050
LR: 0.000825

Epoch 12/40: 100%|██████████| 469/469 [00:19<00:00, 24.47it/s,
loss=95.91, recon=52.03, kld=14.63]

Epoch 12/40:

Total Loss: 99.4559
Recon Loss: 54.6583
KLD Loss: 14.9325
LR: 0.000794

Epoch 13/40: 100%|██████████| 469/469 [00:18<00:00, 25.57it/s,
loss=101.76, recon=54.35, kld=15.80]

Epoch 13/40:

Total Loss: 98.8022
Recon Loss: 53.9328
KLD Loss: 14.9565
LR: 0.000761

Epoch 14/40: 100%|██████████| 469/469 [00:18<00:00, 24.88it/s,
loss=98.64, recon=51.97, kld=15.56]

Epoch 14/40:

Total Loss: 98.1457
Recon Loss: 53.3297
KLD Loss: 14.9387
LR: 0.000727

Epoch 15/40: 100%|██████████| 469/469 [00:18<00:00, 25.15it/s,
loss=96.62, recon=52.07, kld=14.85]

Epoch 15/40:

Total Loss: 97.6092
Recon Loss: 52.7051
KLD Loss: 14.9681
LR: 0.000691

Epoch 16/40: 100%|██████████| 469/469 [00:18<00:00, 24.98it/s, loss=95.13, recon=51.61, kld=14.51]

Epoch 16/40:
Total Loss: 96.9360
Recon Loss: 52.0761
KLD Loss: 14.9533
LR: 0.000655

Epoch 17/40: 100%|██████████| 469/469 [00:19<00:00, 24.66it/s, loss=94.48, recon=50.21, kld=14.76]

Epoch 17/40:
Total Loss: 96.5091
Recon Loss: 51.6168
KLD Loss: 14.9641
LR: 0.000617

Epoch 18/40: 100%|██████████| 469/469 [00:18<00:00, 25.81it/s, loss=97.95, recon=49.97, kld=15.99]

Epoch 18/40:
Total Loss: 95.9897
Recon Loss: 50.9633
KLD Loss: 15.0088
LR: 0.000578

Epoch 19/40: 100%|██████████| 469/469 [00:19<00:00, 24.29it/s, loss=96.47, recon=53.79, kld=14.23]

Epoch 19/40:
Total Loss: 95.5532
Recon Loss: 50.5769
KLD Loss: 14.9921
LR: 0.000539

Epoch 20/40: 100%|██████████| 469/469 [00:18<00:00, 25.14it/s, loss=93.68, recon=49.55, kld=14.71]

Epoch 20/40:
Total Loss: 95.0513
Recon Loss: 50.0424
KLD Loss: 15.0030
LR: 0.000500

Epoch 21/40: 100%|██████████| 469/469 [00:19<00:00, 23.85it/s, loss=92.75, recon=48.88, kld=14.62]

Epoch 21/40:
Total Loss: 94.6754
Recon Loss: 49.5469

KLD Loss: 15.0428
LR: 0.000461

Epoch 22/40: 100%|██████████| 469/469 [00:18<00:00, 25.48it/s,
loss=94.21, recon=48.35, kld=15.29]

Epoch 22/40:
Total Loss: 94.1344
Recon Loss: 49.1056
KLD Loss: 15.0096
LR: 0.000422

Epoch 23/40: 100%|██████████| 469/469 [00:19<00:00, 24.26it/s,
loss=94.39, recon=47.95, kld=15.48]

Epoch 23/40:
Total Loss: 93.9225
Recon Loss: 48.6317
KLD Loss: 15.0969
LR: 0.000383

Epoch 24/40: 100%|██████████| 469/469 [00:18<00:00, 25.39it/s,
loss=89.03, recon=43.53, kld=15.17]

Epoch 24/40:
Total Loss: 93.5247
Recon Loss: 48.2946
KLD Loss: 15.0767
LR: 0.000345

Epoch 25/40: 100%|██████████| 469/469 [00:18<00:00, 24.74it/s,
loss=95.01, recon=48.46, kld=15.52]

Epoch 25/40:
Total Loss: 93.1686
Recon Loss: 47.9342
KLD Loss: 15.0781
LR: 0.000309

Epoch 26/40: 100%|██████████| 469/469 [00:18<00:00, 25.04it/s,
loss=94.28, recon=48.69, kld=15.20]

Epoch 26/40:
Total Loss: 92.7951
Recon Loss: 47.5225
KLD Loss: 15.0909
LR: 0.000273

Epoch 27/40: 100%|██████████| 469/469 [00:18<00:00, 25.42it/s,
loss=91.49, recon=45.74, kld=15.25]

Epoch 27/40:

Total Loss: 92.3580
Recon Loss: 47.1307
KLD Loss: 15.0758
LR: 0.000239

Epoch 28/40: 100%|██████████| 469/469 [00:19<00:00, 24.37it/s, loss=89.46, recon=45.17, kld=14.76]

Epoch 28/40:

Total Loss: 92.1996
Recon Loss: 46.8035
KLD Loss: 15.1320
LR: 0.000206

Epoch 29/40: 100%|██████████| 469/469 [00:18<00:00, 25.26it/s, loss=90.87, recon=46.82, kld=14.68]

Epoch 29/40:

Total Loss: 91.8427
Recon Loss: 46.4020
KLD Loss: 15.1469
LR: 0.000175

Epoch 30/40: 100%|██████████| 469/469 [00:19<00:00, 24.18it/s, loss=93.77, recon=48.49, kld=15.10]

Epoch 30/40:

Total Loss: 91.4986
Recon Loss: 46.1001
KLD Loss: 15.1328
LR: 0.000146

Epoch 31/40: 100%|██████████| 469/469 [00:18<00:00, 25.43it/s, loss=91.13, recon=46.20, kld=14.98]

Epoch 31/40:

Total Loss: 91.3605
Recon Loss: 45.8734
KLD Loss: 15.1624
LR: 0.000120

Epoch 32/40: 100%|██████████| 469/469 [00:19<00:00, 24.31it/s, loss=93.32, recon=46.88, kld=15.48]

Epoch 32/40:

Total Loss: 91.0784
Recon Loss: 45.6107
KLD Loss: 15.1559
LR: 0.000095

Epoch 33/40: 100%|██████████| 469/469 [00:18<00:00, 25.26it/s, loss=98.39, recon=50.43, kld=15.99]

Epoch 33/40:
Total Loss: 90.8734
Recon Loss: 45.3770
KLD Loss: 15.1655
LR: 0.000074

Epoch 34/40: 100%|██████████| 469/469 [00:19<00:00, 24.40it/s, loss=89.45, recon=45.69, kld=14.59]

Epoch 34/40:
Total Loss: 90.6467
Recon Loss: 45.1696
KLD Loss: 15.1591
LR: 0.000054

Epoch 35/40: 100%|██████████| 469/469 [00:18<00:00, 25.69it/s, loss=92.29, recon=47.16, kld=15.04]

Epoch 35/40:
Total Loss: 90.5557
Recon Loss: 44.9765
KLD Loss: 15.1930
LR: 0.000038

Epoch 36/40: 100%|██████████| 469/469 [00:18<00:00, 25.03it/s, loss=89.14, recon=45.10, kld=14.68]

Epoch 36/40:
Total Loss: 90.4626
Recon Loss: 44.8390
KLD Loss: 15.2078
LR: 0.000024

Epoch 37/40: 100%|██████████| 469/469 [00:18<00:00, 24.71it/s, loss=91.32, recon=45.61, kld=15.23]

Epoch 37/40:
Total Loss: 90.3155
Recon Loss: 44.7374
KLD Loss: 15.1927
LR: 0.000014

Epoch 38/40: 100%|██████████| 469/469 [00:18<00:00, 25.45it/s, loss=91.37, recon=44.83, kld=15.51]

Epoch 38/40:
Total Loss: 90.2552
Recon Loss: 44.6624

KLD Loss: 15.1976
LR: 0.000006

Epoch 39/40: 100%|██████████| 469/469 [00:19<00:00, 23.96it/s,
loss=90.50, recon=44.24, kld=15.42]

Epoch 39/40:
Total Loss: 90.1127
Recon Loss: 44.5159
KLD Loss: 15.1989
LR: 0.000002

Epoch 40/40: 100%|██████████| 469/469 [00:18<00:00, 25.32it/s,
loss=91.18, recon=44.67, kld=15.50]

Epoch 40/40:
Total Loss: 90.1548
Recon Loss: 44.6085
KLD Loss: 15.1821
LR: 0.000000

✓ VAE checkpoint saved!

Visualize Reconstruction

```
print("="*60)
print("Visualizing VAE Reconstruction")
print("="*60)

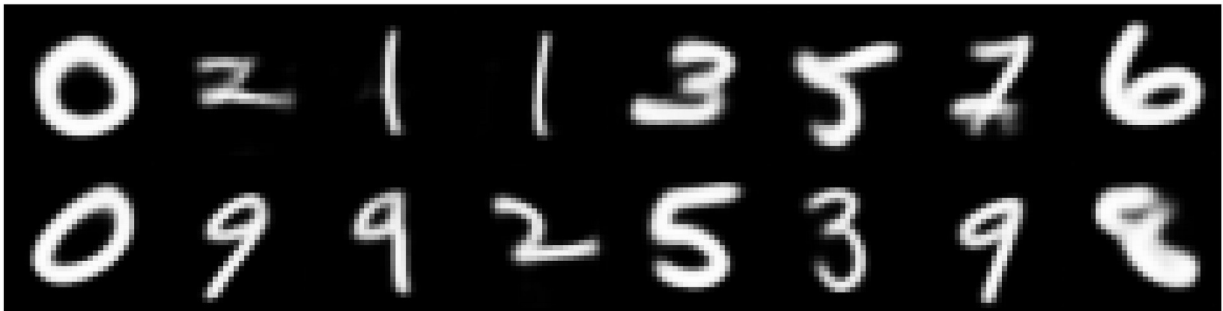
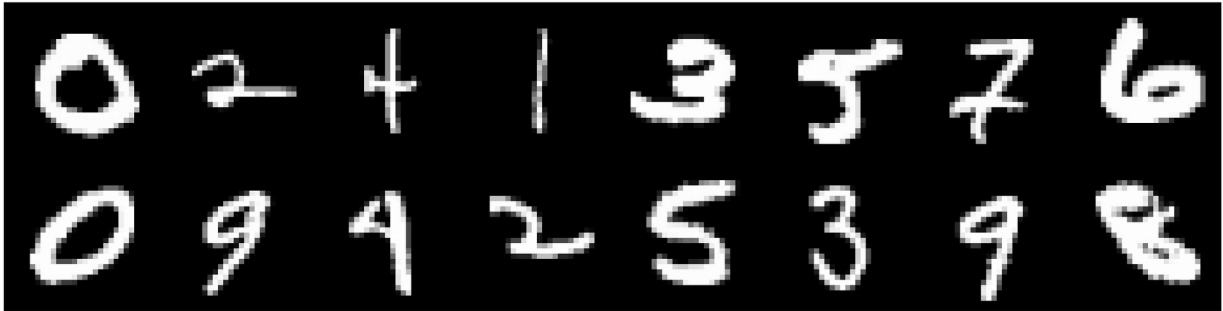
vae.eval()
with torch.no_grad():
    test_dataset = datasets.MNIST('./data', train=False,
    download=True, transform=transform)
    test_loader = DataLoader(test_dataset, batch_size=16,
    shuffle=True)

    data, _ = next(iter(test_loader))
    data = data.to(device)

    recon, mu, logvar = vae(data)

    plot_results(data, nrow=8)
    plot_results(recon, nrow=8)
```

```
=====
Visualizing VAE Reconstruction
=====
```



Visualize Generation

```
print("="*60)
print("Visualizing Random Generation from Latent Space")
print("="*60)

vae.eval()
with torch.no_grad():
    n_samples = 16
    z = torch.randn(n_samples, latent_dim).to(device)

    generated = vae.decode(z)
    generated_np = generated.cpu().numpy()

    plot_results(generated, nrow=8)
```

```
=====
Visualizing Random Generation from Latent Space
=====
```



β -VAE Experiments

```
# Define the  $\beta$  values you would like to explore
latent_dim = 128 # ensure consistency with the base experiment
beta_values = [0.5, 1.0, 3.0, 5.0]

# Containers for results recorded during the experiments
experiment_histories = {}
experiment_summaries = {}

# Prepare a test loader (shared across experiments)
test_dataset = datasets.MNIST('./data', train=False, download=True,
transform=transform)
test_loader = DataLoader(test_dataset, batch_size=32, shuffle=True)

for beta in beta_values:
    print("=" * 60)
    print(f" $\beta$ -VAE Experiment ( $\beta$ = $\beta$ )")
    print("=" * 60)

    vae = ConvVAE(latent_dim=latent_dim, img_channels=1)
    vae, history = train_conv_vae(vae, train_loader, epochs=40,
device=device, beta=beta, lr=1e-3)

    experiment_histories[beta] = history

    final_recon = history['recon'][-1]
    final_kld = history['kld'][-1]
    ratio = final_recon / (final_kld + 1e-8)

    print(f"Final Reconstruction Loss: {final_recon:.4f}")
    print(f"Final KL Divergence: {final_kld:.4f}")
    print(f"Recon/KL Ratio: {ratio:.2f}")

    vae.eval()
    with torch.no_grad():
        batch, _ = next(iter(test_loader))
        batch = batch.to(device)
        recon_batch, _, _ = vae(batch)
```

```

        plot_results(batch[:16], nrow=8, title=f' $\beta$ = $\beta$  • Original
Samples')
        plot_results(recon_batch[:16], nrow=8, title=f' $\beta$ = $\beta$  •
Reconstructions')

        z = torch.randn(32, latent_dim).to(device)
        generated = vae.decode(z)
        plot_results(generated[:16], nrow=8, title=f' $\beta$ = $\beta$  • Random
Generations')

    experiment_summaries[beta] = {
        'final_recon': final_recon,
        'final_kld': final_kld,
        'ratio': ratio,
    }

print("Experiment summaries:")
for beta, summary in experiment_summaries.items():
    print(f' $\beta$ = $\beta$ : Recon={summary['final_recon']:.4f},
KL={summary['final_kld']:.4f}, Recon/KL={summary['ratio']:.2f}")

```

```

=====
 $\beta$ -VAE Experiment ( $\beta$ =0.5)
=====

```

Training Convolutional VAE...

Model parameters: 1,780,545

Epoch 1/40: 100%|██████████| 469/469 [00:19<00:00, 24.54it/s,
loss=59.72, recon=34.22, kld=51.00]

Epoch 1/40:
 Total Loss: 196.1081
 Recon Loss: 66.4895
 KLD Loss: 259.2372
 LR: 0.000998

Epoch 2/40: 100%|██████████| 469/469 [00:17<00:00, 26.26it/s,
loss=50.23, recon=26.88, kld=46.71]

Epoch 2/40:
 Total Loss: 56.0951
 Recon Loss: 31.6140
 KLD Loss: 48.9621
 LR: 0.000994

Epoch 3/40: 100%|██████████| 469/469 [00:18<00:00, 25.34it/s,
loss=48.92, recon=26.34, kld=45.16]

Epoch 3/40:
 Total Loss: 50.9387
 Recon Loss: 27.3731

KLD Loss: 47.1312
LR: 0.000986

Epoch 4/40: 100%|██████████| 469/469 [00:17<00:00, 26.95it/s,
loss=46.52, recon=23.52, kld=45.99]

Epoch 4/40:
Total Loss: 48.8034
Recon Loss: 25.6070
KLD Loss: 46.3929
LR: 0.000976

Epoch 5/40: 100%|██████████| 469/469 [00:17<00:00, 27.41it/s,
loss=46.95, recon=24.60, kld=44.70]

Epoch 5/40:
Total Loss: 47.3981
Recon Loss: 24.4602
KLD Loss: 45.8757
LR: 0.000962

Epoch 6/40: 100%|██████████| 469/469 [00:18<00:00, 25.97it/s,
loss=44.07, recon=22.20, kld=43.74]

Epoch 6/40:
Total Loss: 46.4420
Recon Loss: 23.6866
KLD Loss: 45.5108
LR: 0.000946

Epoch 7/40: 100%|██████████| 469/469 [00:17<00:00, 27.54it/s,
loss=47.19, recon=24.03, kld=46.33]

Epoch 7/40:
Total Loss: 45.6521
Recon Loss: 23.1069
KLD Loss: 45.0903
LR: 0.000926

Epoch 8/40: 100%|██████████| 469/469 [00:16<00:00, 27.64it/s,
loss=44.69, recon=21.90, kld=45.58]

Epoch 8/40:
Total Loss: 44.9449
Recon Loss: 22.5437
KLD Loss: 44.8024
LR: 0.000905

Epoch 9/40: 100%|██████████| 469/469 [00:18<00:00, 26.02it/s,
loss=44.17, recon=22.05, kld=44.23]

Epoch 9/40:

Total Loss: 44.3500
Recon Loss: 22.0731
KLD Loss: 44.5538
LR: 0.000880

Epoch 10/40: 100%|██████████| 469/469 [00:17<00:00, 27.28it/s,
loss=41.36, recon=19.49, kld=43.75]

Epoch 10/40:

Total Loss: 43.8768
Recon Loss: 21.7168
KLD Loss: 44.3200
LR: 0.000854

Epoch 11/40: 100%|██████████| 469/469 [00:18<00:00, 25.70it/s,
loss=43.55, recon=21.19, kld=44.71]

Epoch 11/40:

Total Loss: 43.3351
Recon Loss: 21.3193
KLD Loss: 44.0316
LR: 0.000825

Epoch 12/40: 100%|██████████| 469/469 [00:19<00:00, 24.37it/s,
loss=42.10, recon=20.68, kld=42.84]

Epoch 12/40:

Total Loss: 42.9580
Recon Loss: 21.0246
KLD Loss: 43.8669
LR: 0.000794

Epoch 13/40: 100%|██████████| 469/469 [00:18<00:00, 25.62it/s,
loss=42.89, recon=21.53, kld=42.71]

Epoch 13/40:

Total Loss: 42.5454
Recon Loss: 20.7223
KLD Loss: 43.6462
LR: 0.000761

Epoch 14/40: 100%|██████████| 469/469 [00:19<00:00, 24.07it/s,
loss=40.69, recon=19.32, kld=42.74]

Epoch 14/40:

Total Loss: 42.1988
Recon Loss: 20.4523
KLD Loss: 43.4929
LR: 0.000727

Epoch 15/40: 100%|██████████| 469/469 [00:18<00:00, 25.66it/s, loss=41.00, recon=19.29, kld=43.43]

Epoch 15/40:

Total Loss: 41.8179
Recon Loss: 20.1544
KLD Loss: 43.3271
LR: 0.000691

Epoch 16/40: 100%|██████████| 469/469 [00:18<00:00, 25.16it/s, loss=41.41, recon=20.16, kld=42.49]

Epoch 16/40:

Total Loss: 41.5077
Recon Loss: 19.9103
KLD Loss: 43.1950
LR: 0.000655

Epoch 17/40: 100%|██████████| 469/469 [00:17<00:00, 26.61it/s, loss=43.27, recon=21.13, kld=44.27]

Epoch 17/40:

Total Loss: 41.1976
Recon Loss: 19.7171
KLD Loss: 42.9609
LR: 0.000617

Epoch 18/40: 100%|██████████| 469/469 [00:18<00:00, 25.57it/s, loss=42.00, recon=20.02, kld=43.96]

Epoch 18/40:

Total Loss: 40.8859
Recon Loss: 19.4370
KLD Loss: 42.8978
LR: 0.000578

Epoch 19/40: 100%|██████████| 469/469 [00:18<00:00, 26.00it/s, loss=40.95, recon=19.70, kld=42.50]

Epoch 19/40:

Total Loss: 40.6067
Recon Loss: 19.2381
KLD Loss: 42.7373
LR: 0.000539

Epoch 20/40: 100%|██████████| 469/469 [00:17<00:00, 26.47it/s, loss=40.76, recon=18.97, kld=43.57]

Epoch 20/40:

Total Loss: 40.3329
Recon Loss: 19.0329

KLD Loss: 42.6001
LR: 0.000500

Epoch 21/40: 100%|██████████| 469/469 [00:18<00:00, 25.18it/s,
loss=40.61, recon=19.41, kld=42.40]

Epoch 21/40:
Total Loss: 40.0412
Recon Loss: 18.8036
KLD Loss: 42.4753
LR: 0.000461

Epoch 22/40: 100%|██████████| 469/469 [00:17<00:00, 26.35it/s,
loss=39.57, recon=18.61, kld=41.92]

Epoch 22/40:
Total Loss: 39.8165
Recon Loss: 18.6082
KLD Loss: 42.4166
LR: 0.000422

Epoch 23/40: 100%|██████████| 469/469 [00:19<00:00, 24.50it/s,
loss=41.30, recon=19.20, kld=44.20]

Epoch 23/40:
Total Loss: 39.5486
Recon Loss: 18.4193
KLD Loss: 42.2585
LR: 0.000383

Epoch 24/40: 100%|██████████| 469/469 [00:18<00:00, 25.25it/s,
loss=38.25, recon=17.17, kld=42.17]

Epoch 24/40:
Total Loss: 39.3056
Recon Loss: 18.2188
KLD Loss: 42.1736
LR: 0.000345

Epoch 25/40: 100%|██████████| 469/469 [00:19<00:00, 24.50it/s,
loss=37.08, recon=16.61, kld=40.93]

Epoch 25/40:
Total Loss: 39.0816
Recon Loss: 18.0472
KLD Loss: 42.0688
LR: 0.000309

Epoch 26/40: 100%|██████████| 469/469 [00:18<00:00, 24.69it/s,
loss=38.79, recon=17.24, kld=43.10]

Epoch 26/40:

Total Loss: 38.8988
Recon Loss: 17.8799
KLD Loss: 42.0378
LR: 0.000273

Epoch 27/40: 100%|██████████| 469/469 [00:19<00:00, 24.45it/s,
loss=38.27, recon=17.74, kld=41.06]

Epoch 27/40:

Total Loss: 38.6798
Recon Loss: 17.7255
KLD Loss: 41.9088
LR: 0.000239

Epoch 28/40: 100%|██████████| 469/469 [00:20<00:00, 23.44it/s,
loss=38.65, recon=17.22, kld=42.86]

Epoch 28/40:

Total Loss: 38.4996
Recon Loss: 17.5683
KLD Loss: 41.8626
LR: 0.000206

Epoch 29/40: 100%|██████████| 469/469 [00:19<00:00, 24.00it/s,
loss=39.37, recon=18.25, kld=42.24]

Epoch 29/40:

Total Loss: 38.3230
Recon Loss: 17.4231
KLD Loss: 41.7998
LR: 0.000175

Epoch 30/40: 100%|██████████| 469/469 [00:19<00:00, 24.57it/s,
loss=38.24, recon=16.99, kld=42.51]

Epoch 30/40:

Total Loss: 38.1572
Recon Loss: 17.2746
KLD Loss: 41.7652
LR: 0.000146

Epoch 31/40: 100%|██████████| 469/469 [00:19<00:00, 24.48it/s,
loss=36.85, recon=16.53, kld=40.65]

Epoch 31/40:

Total Loss: 38.0188
Recon Loss: 17.1632
KLD Loss: 41.7110
LR: 0.000120

Epoch 32/40: 100%|██████████| 469/469 [00:19<00:00, 24.30it/s, loss=37.21, recon=16.45, kld=41.52]

Epoch 32/40:

Total Loss: 37.8700
Recon Loss: 17.0373
KLD Loss: 41.6654
LR: 0.000095

Epoch 33/40: 100%|██████████| 469/469 [00:18<00:00, 25.74it/s, loss=38.75, recon=17.29, kld=42.93]

Epoch 33/40:

Total Loss: 37.7751
Recon Loss: 16.9271
KLD Loss: 41.6960
LR: 0.000074

Epoch 34/40: 100%|██████████| 469/469 [00:19<00:00, 24.40it/s, loss=39.32, recon=17.62, kld=43.40]

Epoch 34/40:

Total Loss: 37.6467
Recon Loss: 16.8312
KLD Loss: 41.6310
LR: 0.000054

Epoch 35/40: 100%|██████████| 469/469 [00:18<00:00, 25.59it/s, loss=38.38, recon=17.32, kld=42.11]

Epoch 35/40:

Total Loss: 37.5436
Recon Loss: 16.7460
KLD Loss: 41.5952
LR: 0.000038

Epoch 36/40: 100%|██████████| 469/469 [00:19<00:00, 23.61it/s, loss=37.54, recon=16.74, kld=41.60]

Epoch 36/40:

Total Loss: 37.4875
Recon Loss: 16.6865
KLD Loss: 41.6019
LR: 0.000024

Epoch 37/40: 100%|██████████| 469/469 [00:18<00:00, 24.91it/s, loss=35.04, recon=15.20, kld=39.68]

Epoch 37/40:

Total Loss: 37.4303
Recon Loss: 16.6314

KLD Loss: 41.5979
LR: 0.000014

Epoch 38/40: 100%|██████████| 469/469 [00:19<00:00, 23.81it/s,
loss=37.36, recon=16.56, kld=41.59]

Epoch 38/40:
Total Loss: 37.3780
Recon Loss: 16.5881
KLD Loss: 41.5797
LR: 0.000006

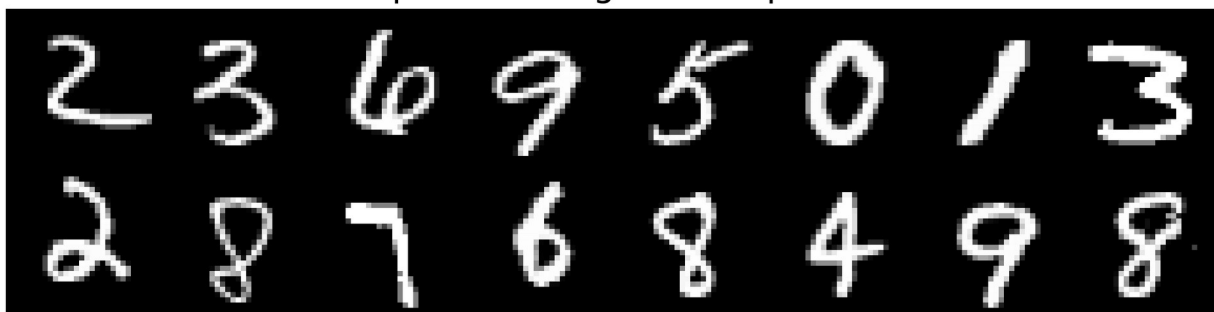
Epoch 39/40: 100%|██████████| 469/469 [00:18<00:00, 24.96it/s,
loss=38.58, recon=17.53, kld=42.11]

Epoch 39/40:
Total Loss: 37.3377
Recon Loss: 16.5555
KLD Loss: 41.5643
LR: 0.000002

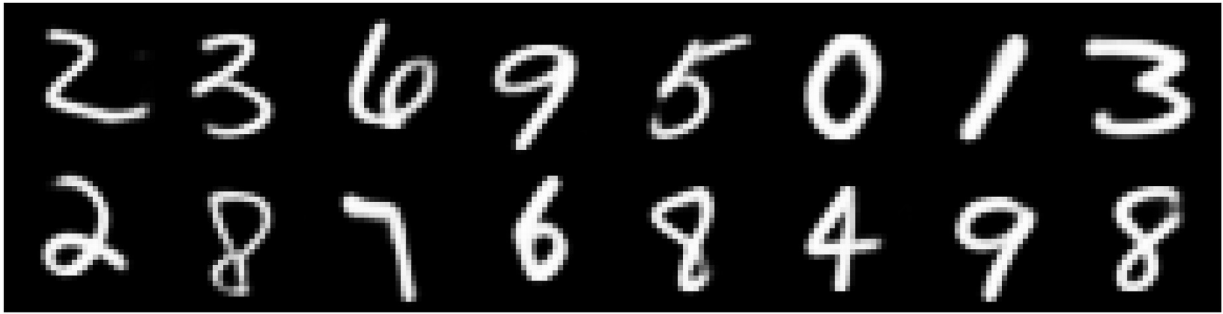
Epoch 40/40: 100%|██████████| 469/469 [00:19<00:00, 23.87it/s,
loss=37.10, recon=16.02, kld=42.16]

Epoch 40/40:
Total Loss: 37.3276
Recon Loss: 16.5292
KLD Loss: 41.5967
LR: 0.000000
Final Reconstruction Loss: 16.5292
Final KL Divergence: 41.5967
Recon/KL Ratio: 0.40

$\beta=0.5$ • Original Samples



$\beta=0.5$ • Reconstructions



$\beta=0.5$ • Random Generations



```
=====
 $\beta$ -VAE Experiment ( $\beta=1.0$ )
=====
Training Convolutional VAE...
Model parameters: 1,780,545

Epoch 1/40: 100%|██████████| 469/469 [00:19<00:00, 24.26it/s,
loss=83.27, recon=50.83, kld=32.44]

Epoch 1/40:
  Total Loss: 182.7860
  Recon Loss: 83.3808
  KLD Loss:   99.4052
  LR:        0.000998

Epoch 2/40: 100%|██████████| 469/469 [00:20<00:00, 22.76it/s,
loss=70.17, recon=37.82, kld=32.35]

Epoch 2/40:
  Total Loss: 75.6954
  Recon Loss: 43.4093
  KLD Loss:   32.2861
  LR:        0.000994

Epoch 3/40: 100%|██████████| 469/469 [00:19<00:00, 24.14it/s,
loss=68.02, recon=37.02, kld=31.01]
```

Epoch 3/40:

Total Loss: 70.1815
Recon Loss: 38.7462
KLD Loss: 31.4353
LR: 0.000986

Epoch 4/40: 100%|██████████| 469/469 [00:19<00:00, 23.50it/s,
loss=67.03, recon=34.50, kld=32.53]

Epoch 4/40:

Total Loss: 67.5875
Recon Loss: 36.5118
KLD Loss: 31.0756
LR: 0.000976

Epoch 5/40: 100%|██████████| 469/469 [00:18<00:00, 24.80it/s,
loss=62.50, recon=31.01, kld=31.50]

Epoch 5/40:

Total Loss: 65.8598
Recon Loss: 35.0302
KLD Loss: 30.8296
LR: 0.000962

Epoch 6/40: 100%|██████████| 469/469 [00:19<00:00, 24.39it/s,
loss=65.10, recon=33.12, kld=31.97]

Epoch 6/40:

Total Loss: 64.5396
Recon Loss: 33.9947
KLD Loss: 30.5450
LR: 0.000946

Epoch 7/40: 100%|██████████| 469/469 [00:18<00:00, 25.07it/s,
loss=63.55, recon=34.03, kld=29.52]

Epoch 7/40:

Total Loss: 63.6356
Recon Loss: 33.2157
KLD Loss: 30.4198
LR: 0.000926

Epoch 8/40: 100%|██████████| 469/469 [00:19<00:00, 24.56it/s,
loss=60.99, recon=30.33, kld=30.66]

Epoch 8/40:

Total Loss: 62.7458
Recon Loss: 32.4713
KLD Loss: 30.2745
LR: 0.000905

Epoch 9/40: 100%|██████████| 469/469 [00:19<00:00, 24.40it/s, loss=63.98, recon=34.61, kld=29.37]

Epoch 9/40:
Total Loss: 62.0096
Recon Loss: 31.8598
KLD Loss: 30.1498
LR: 0.000880

Epoch 10/40: 100%|██████████| 469/469 [00:18<00:00, 25.13it/s, loss=58.32, recon=29.25, kld=29.08]

Epoch 10/40:
Total Loss: 61.3369
Recon Loss: 31.3473
KLD Loss: 29.9896
LR: 0.000854

Epoch 11/40: 100%|██████████| 469/469 [00:19<00:00, 23.67it/s, loss=57.18, recon=27.03, kld=30.14]

Epoch 11/40:
Total Loss: 60.7027
Recon Loss: 30.7948
KLD Loss: 29.9079
LR: 0.000825

Epoch 12/40: 100%|██████████| 469/469 [00:18<00:00, 24.69it/s, loss=59.96, recon=29.32, kld=30.63]

Epoch 12/40:
Total Loss: 60.1857
Recon Loss: 30.3964
KLD Loss: 29.7893
LR: 0.000794

Epoch 13/40: 100%|██████████| 469/469 [00:20<00:00, 23.28it/s, loss=60.08, recon=30.44, kld=29.64]

Epoch 13/40:
Total Loss: 59.6962
Recon Loss: 29.9840
KLD Loss: 29.7122
LR: 0.000761

Epoch 14/40: 100%|██████████| 469/469 [00:18<00:00, 24.77it/s, loss=60.55, recon=30.53, kld=30.02]

Epoch 14/40:
Total Loss: 59.2748
Recon Loss: 29.6842

KLD Loss: 29.5907
LR: 0.000727

Epoch 15/40: 100%|██████████| 469/469 [00:19<00:00, 23.64it/s,
loss=61.22, recon=30.23, kld=30.99]

Epoch 15/40:
Total Loss: 58.7980
Recon Loss: 29.2412
KLD Loss: 29.5567
LR: 0.000691

Epoch 16/40: 100%|██████████| 469/469 [00:18<00:00, 24.96it/s,
loss=62.64, recon=31.82, kld=30.82]

Epoch 16/40:
Total Loss: 58.4099
Recon Loss: 28.9607
KLD Loss: 29.4492
LR: 0.000655

Epoch 17/40: 100%|██████████| 469/469 [00:19<00:00, 23.48it/s,
loss=56.19, recon=27.70, kld=28.50]

Epoch 17/40:
Total Loss: 58.0375
Recon Loss: 28.6726
KLD Loss: 29.3649
LR: 0.000617

Epoch 18/40: 100%|██████████| 469/469 [00:19<00:00, 24.49it/s,
loss=62.30, recon=31.72, kld=30.58]

Epoch 18/40:
Total Loss: 57.6903
Recon Loss: 28.3815
KLD Loss: 29.3089
LR: 0.000578

Epoch 19/40: 100%|██████████| 469/469 [00:20<00:00, 22.93it/s,
loss=59.32, recon=29.20, kld=30.12]

Epoch 19/40:
Total Loss: 57.2981
Recon Loss: 28.0348
KLD Loss: 29.2633
LR: 0.000539

Epoch 20/40: 100%|██████████| 469/469 [00:19<00:00, 24.61it/s,
loss=57.80, recon=27.75, kld=30.05]

Epoch 20/40:

Total Loss: 56.9818
Recon Loss: 27.7662
KLD Loss: 29.2156
LR: 0.000500

Epoch 21/40: 100%|██████████| 469/469 [00:20<00:00, 23.37it/s,
loss=56.89, recon=28.68, kld=28.21]

Epoch 21/40:

Total Loss: 56.6410
Recon Loss: 27.4887
KLD Loss: 29.1523
LR: 0.000461

Epoch 22/40: 100%|██████████| 469/469 [00:19<00:00, 24.51it/s,
loss=56.12, recon=27.15, kld=28.97]

Epoch 22/40:

Total Loss: 56.3819
Recon Loss: 27.2335
KLD Loss: 29.1484
LR: 0.000422

Epoch 23/40: 100%|██████████| 469/469 [00:19<00:00, 24.31it/s,
loss=54.36, recon=24.95, kld=29.41]

Epoch 23/40:

Total Loss: 56.0343
Recon Loss: 26.9653
KLD Loss: 29.0690
LR: 0.000383

Epoch 24/40: 100%|██████████| 469/469 [00:19<00:00, 24.22it/s,
loss=56.13, recon=26.71, kld=29.43]

Epoch 24/40:

Total Loss: 55.7635
Recon Loss: 26.7192
KLD Loss: 29.0444
LR: 0.000345

Epoch 25/40: 100%|██████████| 469/469 [00:19<00:00, 24.62it/s,
loss=54.64, recon=25.99, kld=28.65]

Epoch 25/40:

Total Loss: 55.4822
Recon Loss: 26.4875
KLD Loss: 28.9947
LR: 0.000309

Epoch 26/40: 100%|██████████| 469/469 [00:19<00:00, 23.69it/s, loss=52.23, recon=22.86, kld=29.37]

Epoch 26/40:
Total Loss: 55.2583
Recon Loss: 26.2771
KLD Loss: 28.9811
LR: 0.000273

Epoch 27/40: 100%|██████████| 469/469 [00:19<00:00, 24.24it/s, loss=54.62, recon=25.88, kld=28.74]

Epoch 27/40:
Total Loss: 54.9933
Recon Loss: 26.0304
KLD Loss: 28.9629
LR: 0.000239

Epoch 28/40: 100%|██████████| 469/469 [00:19<00:00, 23.66it/s, loss=56.92, recon=27.72, kld=29.21]

Epoch 28/40:
Total Loss: 54.7428
Recon Loss: 25.8064
KLD Loss: 28.9364
LR: 0.000206

Epoch 29/40: 100%|██████████| 469/469 [00:19<00:00, 24.34it/s, loss=54.55, recon=25.67, kld=28.88]

Epoch 29/40:
Total Loss: 54.5623
Recon Loss: 25.6435
KLD Loss: 28.9189
LR: 0.000175

Epoch 30/40: 100%|██████████| 469/469 [00:20<00:00, 23.41it/s, loss=51.86, recon=24.29, kld=27.56]

Epoch 30/40:
Total Loss: 54.3722
Recon Loss: 25.4572
KLD Loss: 28.9150
LR: 0.000146

Epoch 31/40: 100%|██████████| 469/469 [00:18<00:00, 24.84it/s, loss=54.80, recon=25.21, kld=29.59]

Epoch 31/40:
Total Loss: 54.1797
Recon Loss: 25.2593

KLD Loss: 28.9204
LR: 0.000120

Epoch 32/40: 100%|██████████| 469/469 [00:20<00:00, 23.39it/s,
loss=51.58, recon=23.75, kld=27.83]

Epoch 32/40:
Total Loss: 54.0034
Recon Loss: 25.1110
KLD Loss: 28.8924
LR: 0.000095

Epoch 33/40: 100%|██████████| 469/469 [00:19<00:00, 24.19it/s,
loss=52.59, recon=24.13, kld=28.47]

Epoch 33/40:
Total Loss: 53.8000
Recon Loss: 24.9865
KLD Loss: 28.8135
LR: 0.000074

Epoch 34/40: 100%|██████████| 469/469 [00:19<00:00, 23.52it/s,
loss=53.93, recon=25.14, kld=28.79]

Epoch 34/40:
Total Loss: 53.6922
Recon Loss: 24.8599
KLD Loss: 28.8323
LR: 0.000054

Epoch 35/40: 100%|██████████| 469/469 [00:19<00:00, 24.27it/s,
loss=53.67, recon=24.03, kld=29.64]

Epoch 35/40:
Total Loss: 53.5843
Recon Loss: 24.7349
KLD Loss: 28.8494
LR: 0.000038

Epoch 36/40: 100%|██████████| 469/469 [00:20<00:00, 23.36it/s,
loss=52.53, recon=24.08, kld=28.45]

Epoch 36/40:
Total Loss: 53.4948
Recon Loss: 24.6741
KLD Loss: 28.8207
LR: 0.000024

Epoch 37/40: 100%|██████████| 469/469 [00:19<00:00, 24.47it/s,
loss=49.61, recon=21.94, kld=27.67]

Epoch 37/40:

Total Loss: 53.4157
Recon Loss: 24.5945
KLD Loss: 28.8212
LR: 0.000014

Epoch 38/40: 100%|██████████| 469/469 [00:20<00:00, 23.11it/s,
loss=52.60, recon=23.40, kld=29.20]

Epoch 38/40:

Total Loss: 53.3768
Recon Loss: 24.5457
KLD Loss: 28.8311
LR: 0.000006

Epoch 39/40: 100%|██████████| 469/469 [00:18<00:00, 24.71it/s,
loss=52.81, recon=23.98, kld=28.83]

Epoch 39/40:

Total Loss: 53.3178
Recon Loss: 24.5037
KLD Loss: 28.8141
LR: 0.000002

Epoch 40/40: 100%|██████████| 469/469 [00:19<00:00, 23.51it/s,
loss=52.28, recon=23.74, kld=28.55]

Epoch 40/40:

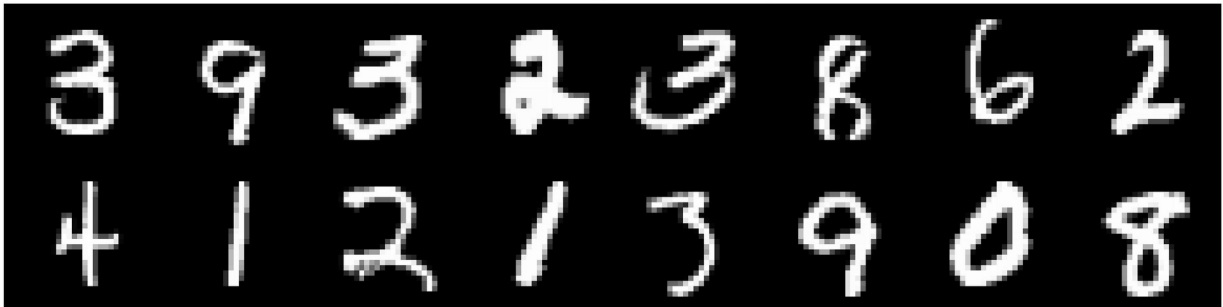
Total Loss: 53.2964
Recon Loss: 24.4601
KLD Loss: 28.8363
LR: 0.000000

Final Reconstruction Loss: 24.4601

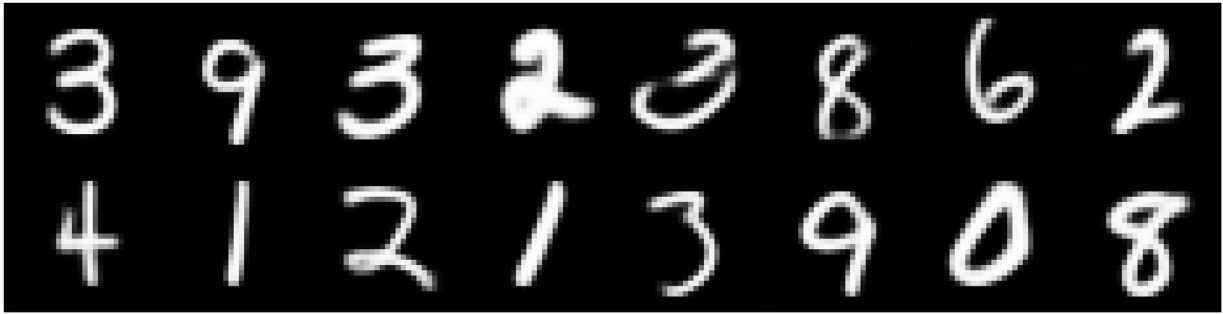
Final KL Divergence: 28.8363

Recon/KL Ratio: 0.85

$\beta=1.0$ • Original Samples



$\beta=1.0$ • Reconstructions



$\beta=1.0$ • Random Generations



```
=====
 $\beta$ -VAE Experiment ( $\beta=3.0$ )
=====
Training Convolutional VAE...
Model parameters: 1,780,545

Epoch 1/40: 100%|██████████| 469/469 [00:19<00:00, 23.52it/s,
loss=128.60, recon=83.14, kld=15.15]

Epoch 1/40:
  Total Loss: 482.0111
  Recon Loss: 121.2041
  KLD Loss:   120.2690
  LR:        0.000998

Epoch 2/40: 100%|██████████| 469/469 [00:20<00:00, 22.41it/s,
loss=117.81, recon=68.97, kld=16.28]

Epoch 2/40:
  Total Loss: 121.6481
  Recon Loss: 77.1755
  KLD Loss:   14.8242
  LR:        0.000994

Epoch 3/40: 100%|██████████| 469/469 [00:19<00:00, 23.69it/s,
loss=114.11, recon=70.86, kld=14.41]
```

Epoch 3/40:

Total Loss: 113.7831
Recon Loss: 69.2616
KLD Loss: 14.8405
LR: 0.000986

Epoch 4/40: 100%|██████████| 469/469 [00:20<00:00, 22.47it/s,
loss=99.59, recon=57.60, kld=14.00]

Epoch 4/40:

Total Loss: 109.9007
Recon Loss: 65.2363
KLD Loss: 14.8881
LR: 0.000976

Epoch 5/40: 100%|██████████| 469/469 [00:19<00:00, 24.11it/s,
loss=103.03, recon=55.48, kld=15.85]

Epoch 5/40:

Total Loss: 107.5803
Recon Loss: 62.8536
KLD Loss: 14.9089
LR: 0.000962

Epoch 6/40: 100%|██████████| 469/469 [00:20<00:00, 22.71it/s,
loss=113.33, recon=66.66, kld=15.56]

Epoch 6/40:

Total Loss: 105.8051
Recon Loss: 60.9887
KLD Loss: 14.9388
LR: 0.000946

Epoch 7/40: 100%|██████████| 469/469 [00:19<00:00, 23.93it/s,
loss=97.83, recon=54.60, kld=14.41]

Epoch 7/40:

Total Loss: 104.3761
Recon Loss: 59.5769
KLD Loss: 14.9331
LR: 0.000926

Epoch 8/40: 100%|██████████| 469/469 [00:20<00:00, 23.06it/s,
loss=106.14, recon=60.97, kld=15.06]

Epoch 8/40:

Total Loss: 103.1039
Recon Loss: 58.2773
KLD Loss: 14.9422
LR: 0.000905

Epoch 9/40: 100%|██████████| 469/469 [00:19<00:00, 24.10it/s, loss=101.66, recon=55.93, kld=15.24]

Epoch 9/40:
Total Loss: 102.1155
Recon Loss: 57.2507
KLD Loss: 14.9549
LR: 0.000880

Epoch 10/40: 100%|██████████| 469/469 [00:20<00:00, 22.72it/s, loss=96.73, recon=52.59, kld=14.71]

Epoch 10/40:
Total Loss: 101.0986
Recon Loss: 56.2680
KLD Loss: 14.9435
LR: 0.000854

Epoch 11/40: 100%|██████████| 469/469 [00:19<00:00, 24.32it/s, loss=95.33, recon=53.66, kld=13.89]

Epoch 11/40:
Total Loss: 100.2726
Recon Loss: 55.4251
KLD Loss: 14.9492
LR: 0.000825

Epoch 12/40: 100%|██████████| 469/469 [00:20<00:00, 22.86it/s, loss=101.35, recon=54.64, kld=15.57]

Epoch 12/40:
Total Loss: 99.5589
Recon Loss: 54.6742
KLD Loss: 14.9616
LR: 0.000794

Epoch 13/40: 100%|██████████| 469/469 [00:19<00:00, 23.91it/s, loss=102.18, recon=57.41, kld=14.92]

Epoch 13/40:
Total Loss: 98.9293
Recon Loss: 54.0288
KLD Loss: 14.9668
LR: 0.000761

Epoch 14/40: 100%|██████████| 469/469 [00:20<00:00, 22.53it/s, loss=95.54, recon=52.88, kld=14.22]

Epoch 14/40:
Total Loss: 98.3379
Recon Loss: 53.4266

KLD Loss: 14.9704
LR: 0.000727

Epoch 15/40: 100%|██████████| 469/469 [00:19<00:00, 23.60it/s,
loss=100.06, recon=55.59, kld=14.82]

Epoch 15/40:
Total Loss: 97.7743
Recon Loss: 52.8645
KLD Loss: 14.9699
LR: 0.000691

Epoch 16/40: 100%|██████████| 469/469 [00:20<00:00, 22.97it/s,
loss=98.02, recon=53.21, kld=14.94]

Epoch 16/40:
Total Loss: 97.2592
Recon Loss: 52.2305
KLD Loss: 15.0096
LR: 0.000655

Epoch 17/40: 100%|██████████| 469/469 [00:19<00:00, 23.58it/s,
loss=94.44, recon=48.83, kld=15.20]

Epoch 17/40:
Total Loss: 96.5811
Recon Loss: 51.6807
KLD Loss: 14.9668
LR: 0.000617

Epoch 18/40: 100%|██████████| 469/469 [00:20<00:00, 22.49it/s,
loss=96.28, recon=50.67, kld=15.20]

Epoch 18/40:
Total Loss: 96.1451
Recon Loss: 51.1757
KLD Loss: 14.9898
LR: 0.000578

Epoch 19/40: 100%|██████████| 469/469 [00:19<00:00, 23.56it/s,
loss=95.60, recon=51.41, kld=14.73]

Epoch 19/40:
Total Loss: 95.6792
Recon Loss: 50.7249
KLD Loss: 14.9848
LR: 0.000539

Epoch 20/40: 100%|██████████| 469/469 [00:20<00:00, 22.57it/s,
loss=98.96, recon=54.55, kld=14.81]

Epoch 20/40:

Total Loss: 95.2071
Recon Loss: 50.2245
KLD Loss: 14.9942
LR: 0.000500

Epoch 21/40: 100%|██████████| 469/469 [00:19<00:00, 24.09it/s,
loss=97.04, recon=49.04, kld=16.00]

Epoch 21/40:

Total Loss: 94.7744
Recon Loss: 49.7776
KLD Loss: 14.9989
LR: 0.000461

Epoch 22/40: 100%|██████████| 469/469 [00:20<00:00, 22.65it/s,
loss=95.01, recon=48.51, kld=15.50]

Epoch 22/40:

Total Loss: 94.3290
Recon Loss: 49.3308
KLD Loss: 14.9994
LR: 0.000422

Epoch 23/40: 100%|██████████| 469/469 [00:19<00:00, 23.48it/s,
loss=95.45, recon=50.37, kld=15.03]

Epoch 23/40:

Total Loss: 93.9825
Recon Loss: 48.8910
KLD Loss: 15.0305
LR: 0.000383

Epoch 24/40: 100%|██████████| 469/469 [00:20<00:00, 22.64it/s,
loss=97.28, recon=50.30, kld=15.66]

Epoch 24/40:

Total Loss: 93.6133
Recon Loss: 48.5261
KLD Loss: 15.0291
LR: 0.000345

Epoch 25/40: 100%|██████████| 469/469 [00:19<00:00, 23.78it/s,
loss=91.44, recon=48.27, kld=14.39]

Epoch 25/40:

Total Loss: 93.2024
Recon Loss: 48.1135
KLD Loss: 15.0296
LR: 0.000309

Epoch 26/40: 100%|██████████| 469/469 [00:20<00:00, 22.45it/s, loss=92.87, recon=47.65, kld=15.07]

Epoch 26/40:
Total Loss: 92.8717
Recon Loss: 47.7923
KLD Loss: 15.0265
LR: 0.000273

Epoch 27/40: 100%|██████████| 469/469 [00:19<00:00, 23.70it/s, loss=92.01, recon=46.95, kld=15.02]

Epoch 27/40:
Total Loss: 92.5908
Recon Loss: 47.3477
KLD Loss: 15.0810
LR: 0.000239

Epoch 28/40: 100%|██████████| 469/469 [00:20<00:00, 22.95it/s, loss=91.01, recon=45.22, kld=15.26]

Epoch 28/40:
Total Loss: 92.1202
Recon Loss: 46.9900
KLD Loss: 15.0434
LR: 0.000206

Epoch 29/40: 100%|██████████| 469/469 [00:19<00:00, 23.58it/s, loss=91.77, recon=46.81, kld=14.99]

Epoch 29/40:
Total Loss: 91.8616
Recon Loss: 46.6508
KLD Loss: 15.0703
LR: 0.000175

Epoch 30/40: 100%|██████████| 469/469 [00:20<00:00, 22.53it/s, loss=90.13, recon=45.92, kld=14.74]

Epoch 30/40:
Total Loss: 91.6113
Recon Loss: 46.3871
KLD Loss: 15.0747
LR: 0.000146

Epoch 31/40: 100%|██████████| 469/469 [00:19<00:00, 24.03it/s, loss=95.25, recon=48.53, kld=15.57]

Epoch 31/40:
Total Loss: 91.3577
Recon Loss: 46.0706

KLD Loss: 15.0957
LR: 0.000120

Epoch 32/40: 100%|██████████| 469/469 [00:20<00:00, 23.07it/s,
loss=86.19, recon=41.98, kld=14.74]

Epoch 32/40:
Total Loss: 91.1374
Recon Loss: 45.8070
KLD Loss: 15.1101
LR: 0.000095

Epoch 33/40: 100%|██████████| 469/469 [00:19<00:00, 23.79it/s,
loss=94.84, recon=48.22, kld=15.54]

Epoch 33/40:
Total Loss: 90.8941
Recon Loss: 45.5827
KLD Loss: 15.1038
LR: 0.000074

Epoch 34/40: 100%|██████████| 469/469 [00:20<00:00, 23.08it/s,
loss=91.28, recon=46.51, kld=14.92]

Epoch 34/40:
Total Loss: 90.7223
Recon Loss: 45.3462
KLD Loss: 15.1253
LR: 0.000054

Epoch 35/40: 100%|██████████| 469/469 [00:20<00:00, 23.36it/s,
loss=90.36, recon=43.72, kld=15.55]

Epoch 35/40:
Total Loss: 90.5834
Recon Loss: 45.1580
KLD Loss: 15.1418
LR: 0.000038

Epoch 36/40: 100%|██████████| 469/469 [00:20<00:00, 22.67it/s,
loss=89.94, recon=45.44, kld=14.83]

Epoch 36/40:
Total Loss: 90.4682
Recon Loss: 45.0670
KLD Loss: 15.1338
LR: 0.000024

Epoch 37/40: 100%|██████████| 469/469 [00:19<00:00, 23.80it/s,
loss=86.78, recon=42.24, kld=14.85]

Epoch 37/40:

Total Loss: 90.3940
Recon Loss: 44.9700
KLD Loss: 15.1413
LR: 0.000014

Epoch 38/40: 100%|██████████| 469/469 [00:20<00:00, 23.01it/s,
loss=86.46, recon=41.94, kld=14.84]

Epoch 38/40:

Total Loss: 90.2449
Recon Loss: 44.8420
KLD Loss: 15.1343
LR: 0.000006

Epoch 39/40: 100%|██████████| 469/469 [00:19<00:00, 23.75it/s,
loss=89.70, recon=45.41, kld=14.76]

Epoch 39/40:

Total Loss: 90.1992
Recon Loss: 44.7549
KLD Loss: 15.1481
LR: 0.000002

Epoch 40/40: 100%|██████████| 469/469 [00:20<00:00, 23.14it/s,
loss=89.55, recon=44.83, kld=14.91]

Epoch 40/40:

Total Loss: 90.2523
Recon Loss: 44.7836
KLD Loss: 15.1563
LR: 0.000000

Final Reconstruction Loss: 44.7836

Final KL Divergence: 15.1563

Recon/KL Ratio: 2.95

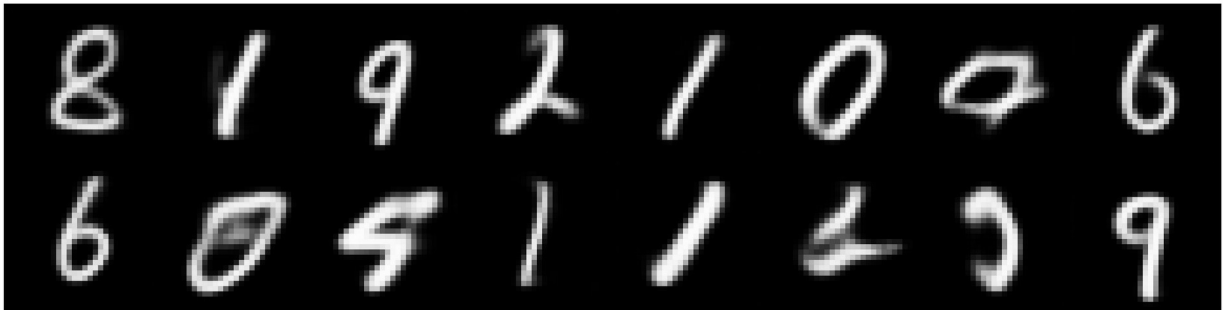
$\beta=3.0$ • Original Samples



$\beta=3.0$ • Reconstructions



$\beta=3.0$ • Random Generations



```
=====
 $\beta$ -VAE Experiment ( $\beta=5.0$ )
=====
Training Convolutional VAE...
Model parameters: 1,780,545

Epoch 1/40: 100%|██████████| 469/469 [00:20<00:00, 23.06it/s,
loss=161.13, recon=117.51, kld=8.72]

Epoch 1/40:
  Total Loss: 1116.0322
  Recon Loss: 148.2675
  KLD Loss:   193.5529
  LR:        0.000998

Epoch 2/40: 100%|██████████| 469/469 [00:21<00:00, 21.93it/s,
loss=148.55, recon=98.40, kld=10.03]

Epoch 2/40:
  Total Loss: 149.3252
  Recon Loss: 103.0105
  KLD Loss:   9.2629
  LR:        0.000994

Epoch 3/40: 100%|██████████| 469/469 [00:20<00:00, 23.36it/s,
loss=135.05, recon=86.54, kld=9.70]
```

Epoch 3/40:

Total Loss: 139.3262
Recon Loss: 91.5998
KLD Loss: 9.5453
LR: 0.000986

Epoch 4/40: 100%|██████████| 469/469 [00:21<00:00, 22.23it/s,
loss=136.15, recon=87.54, kld=9.72]

Epoch 4/40:

Total Loss: 135.1728
Recon Loss: 86.4366
KLD Loss: 9.7472
LR: 0.000976

Epoch 5/40: 100%|██████████| 469/469 [00:20<00:00, 23.28it/s,
loss=133.13, recon=82.97, kld=10.03]

Epoch 5/40:

Total Loss: 132.5186
Recon Loss: 83.1866
KLD Loss: 9.8664
LR: 0.000962

Epoch 6/40: 100%|██████████| 469/469 [00:21<00:00, 22.09it/s,
loss=126.07, recon=75.75, kld=10.06]

Epoch 6/40:

Total Loss: 130.6595
Recon Loss: 80.6402
KLD Loss: 10.0039
LR: 0.000946

Epoch 7/40: 100%|██████████| 469/469 [00:20<00:00, 23.21it/s,
loss=126.26, recon=75.53, kld=10.15]

Epoch 7/40:

Total Loss: 129.2405
Recon Loss: 78.8595
KLD Loss: 10.0762
LR: 0.000926

Epoch 8/40: 100%|██████████| 469/469 [00:21<00:00, 22.19it/s,
loss=121.49, recon=71.69, kld=9.96]

Epoch 8/40:

Total Loss: 128.0062
Recon Loss: 77.3116
KLD Loss: 10.1389
LR: 0.000905

Epoch 9/40: 100%|██████████| 469/469 [00:20<00:00, 23.18it/s, loss=130.80, recon=80.12, kld=10.14]

Epoch 9/40:
Total Loss: 127.0008
Recon Loss: 76.0378
KLD Loss: 10.1926
LR: 0.000880

Epoch 10/40: 100%|██████████| 469/469 [00:21<00:00, 21.69it/s, loss=129.20, recon=76.45, kld=10.55]

Epoch 10/40:
Total Loss: 126.1158
Recon Loss: 74.8931
KLD Loss: 10.2445
LR: 0.000854

Epoch 11/40: 100%|██████████| 469/469 [00:20<00:00, 22.70it/s, loss=122.82, recon=72.69, kld=10.03]

Epoch 11/40:
Total Loss: 125.2441
Recon Loss: 73.7758
KLD Loss: 10.2937
LR: 0.000825

Epoch 12/40: 100%|██████████| 469/469 [00:21<00:00, 21.52it/s, loss=123.69, recon=71.02, kld=10.53]

Epoch 12/40:
Total Loss: 124.6411
Recon Loss: 72.8841
KLD Loss: 10.3514
LR: 0.000794

Epoch 13/40: 100%|██████████| 469/469 [00:20<00:00, 22.41it/s, loss=119.06, recon=67.61, kld=10.29]

Epoch 13/40:
Total Loss: 123.8999
Recon Loss: 72.0736
KLD Loss: 10.3653
LR: 0.000761

Epoch 14/40: 100%|██████████| 469/469 [00:22<00:00, 21.14it/s, loss=125.28, recon=75.02, kld=10.05]

Epoch 14/40:
Total Loss: 123.3851
Recon Loss: 71.3658

KLD Loss: 10.4039
LR: 0.000727

Epoch 15/40: 100%|██████████| 469/469 [00:21<00:00, 22.12it/s,
loss=116.69, recon=65.94, kld=10.15]

Epoch 15/40:
Total Loss: 122.8177
Recon Loss: 70.5690
KLD Loss: 10.4498
LR: 0.000691

Epoch 16/40: 100%|██████████| 469/469 [00:21<00:00, 21.95it/s,
loss=125.73, recon=74.39, kld=10.27]

Epoch 16/40:
Total Loss: 122.3328
Recon Loss: 70.0138
KLD Loss: 10.4638
LR: 0.000655

Epoch 17/40: 100%|██████████| 469/469 [00:21<00:00, 21.50it/s,
loss=124.12, recon=71.06, kld=10.61]

Epoch 17/40:
Total Loss: 121.6562
Recon Loss: 69.1845
KLD Loss: 10.4944
LR: 0.000617

Epoch 18/40: 100%|██████████| 469/469 [00:21<00:00, 22.30it/s,
loss=126.30, recon=72.85, kld=10.69]

Epoch 18/40:
Total Loss: 121.3150
Recon Loss: 68.7168
KLD Loss: 10.5197
LR: 0.000578

Epoch 19/40: 100%|██████████| 469/469 [00:22<00:00, 21.13it/s,
loss=123.30, recon=67.51, kld=11.16]

Epoch 19/40:
Total Loss: 120.8541
Recon Loss: 68.1908
KLD Loss: 10.5327
LR: 0.000539

Epoch 20/40: 100%|██████████| 469/469 [00:21<00:00, 22.20it/s,
loss=110.70, recon=60.32, kld=10.08]

Epoch 20/40:

Total Loss: 120.4143
Recon Loss: 67.4362
KLD Loss: 10.5956
LR: 0.000500

Epoch 21/40: 100%|██████████| 469/469 [00:21<00:00, 21.34it/s,
loss=125.76, recon=71.84, kld=10.78]

Epoch 21/40:

Total Loss: 119.8943
Recon Loss: 66.9486
KLD Loss: 10.5891
LR: 0.000461

Epoch 22/40: 100%|██████████| 469/469 [00:20<00:00, 22.51it/s,
loss=118.71, recon=66.50, kld=10.44]

Epoch 22/40:

Total Loss: 119.7036
Recon Loss: 66.5271
KLD Loss: 10.6353
LR: 0.000422

Epoch 23/40: 100%|██████████| 469/469 [00:22<00:00, 21.19it/s,
loss=120.65, recon=66.79, kld=10.77]

Epoch 23/40:

Total Loss: 119.2340
Recon Loss: 65.8607
KLD Loss: 10.6747
LR: 0.000383

Epoch 24/40: 100%|██████████| 469/469 [00:21<00:00, 22.17it/s,
loss=118.23, recon=64.85, kld=10.68]

Epoch 24/40:

Total Loss: 118.7420
Recon Loss: 65.3962
KLD Loss: 10.6692
LR: 0.000345

Epoch 25/40: 100%|██████████| 469/469 [00:22<00:00, 21.27it/s,
loss=113.04, recon=61.24, kld=10.36]

Epoch 25/40:

Total Loss: 118.4131
Recon Loss: 64.9100
KLD Loss: 10.7006
LR: 0.000309

Epoch 26/40: 100%|██████████| 469/469 [00:21<00:00, 21.45it/s, loss=117.99, recon=64.20, kld=10.76]

Epoch 26/40:
Total Loss: 118.0768
Recon Loss: 64.5025
KLD Loss: 10.7149
LR: 0.000273

Epoch 27/40: 100%|██████████| 469/469 [00:21<00:00, 22.31it/s, loss=118.74, recon=65.79, kld=10.59]

Epoch 27/40:
Total Loss: 117.9612
Recon Loss: 63.9831
KLD Loss: 10.7956
LR: 0.000239

Epoch 28/40: 100%|██████████| 469/469 [00:21<00:00, 21.40it/s, loss=121.49, recon=66.53, kld=10.99]

Epoch 28/40:
Total Loss: 117.4315
Recon Loss: 63.6095
KLD Loss: 10.7644
LR: 0.000206

Epoch 29/40: 100%|██████████| 469/469 [00:20<00:00, 22.61it/s, loss=112.93, recon=59.33, kld=10.72]

Epoch 29/40:
Total Loss: 117.0950
Recon Loss: 63.1906
KLD Loss: 10.7809
LR: 0.000175

Epoch 30/40: 100%|██████████| 469/469 [00:21<00:00, 21.67it/s, loss=111.71, recon=58.69, kld=10.60]

Epoch 30/40:
Total Loss: 116.8403
Recon Loss: 62.8418
KLD Loss: 10.7997
LR: 0.000146

Epoch 31/40: 100%|██████████| 469/469 [00:21<00:00, 22.23it/s, loss=114.44, recon=59.87, kld=10.91]

Epoch 31/40:
Total Loss: 116.5725
Recon Loss: 62.4610

KLD Loss: 10.8223
LR: 0.000120

Epoch 32/40: 100%|██████████| 469/469 [00:22<00:00, 20.69it/s,
loss=114.56, recon=59.49, kld=11.01]

Epoch 32/40:
Total Loss: 116.3949
Recon Loss: 62.1800
KLD Loss: 10.8430
LR: 0.000095

Epoch 33/40: 100%|██████████| 469/469 [00:21<00:00, 22.23it/s,
loss=118.31, recon=64.17, kld=10.83]

Epoch 33/40:
Total Loss: 116.2041
Recon Loss: 61.8518
KLD Loss: 10.8705
LR: 0.000074

Epoch 34/40: 100%|██████████| 469/469 [00:22<00:00, 20.84it/s,
loss=116.42, recon=62.25, kld=10.83]

Epoch 34/40:
Total Loss: 115.9903
Recon Loss: 61.6276
KLD Loss: 10.8725
LR: 0.000054

Epoch 35/40: 100%|██████████| 469/469 [00:21<00:00, 21.83it/s,
loss=116.28, recon=60.13, kld=11.23]

Epoch 35/40:
Total Loss: 115.8868
Recon Loss: 61.4206
KLD Loss: 10.8932
LR: 0.000038

Epoch 36/40: 100%|██████████| 469/469 [00:21<00:00, 22.11it/s,
loss=117.45, recon=62.22, kld=11.04]

Epoch 36/40:
Total Loss: 115.6900
Recon Loss: 61.2864
KLD Loss: 10.8807
LR: 0.000024

Epoch 37/40: 100%|██████████| 469/469 [00:22<00:00, 20.74it/s,
loss=117.44, recon=64.26, kld=10.64]

Epoch 37/40:

Total Loss: 115.4650
Recon Loss: 61.0083
KLD Loss: 10.8914
LR: 0.000014

Epoch 38/40: 100%|██████████| 469/469 [00:21<00:00, 21.98it/s,
loss=114.42, recon=61.15, kld=10.65]

Epoch 38/40:

Total Loss: 115.5363
Recon Loss: 61.0420
KLD Loss: 10.8989
LR: 0.000006

Epoch 39/40: 100%|██████████| 469/469 [00:22<00:00, 20.88it/s,
loss=116.07, recon=61.58, kld=10.90]

Epoch 39/40:

Total Loss: 115.3587
Recon Loss: 60.8814
KLD Loss: 10.8954
LR: 0.000002

Epoch 40/40: 100%|██████████| 469/469 [00:21<00:00, 21.73it/s,
loss=111.78, recon=56.76, kld=11.00]

Epoch 40/40:

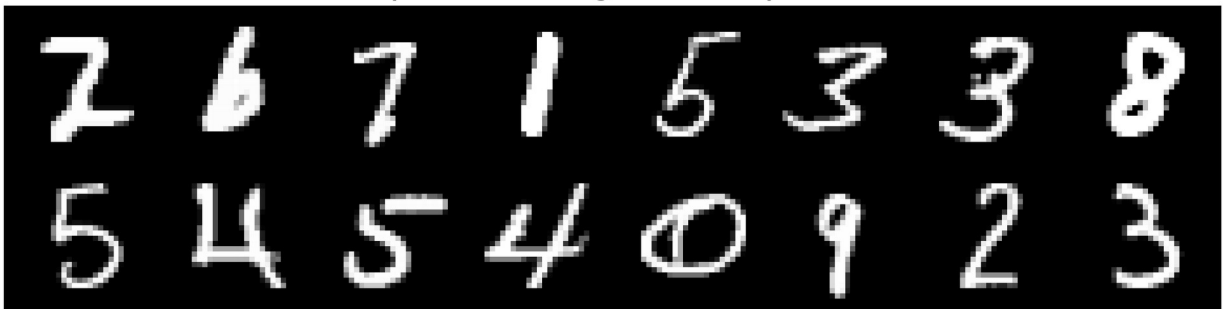
Total Loss: 115.4149
Recon Loss: 60.8821
KLD Loss: 10.9066
LR: 0.000000

Final Reconstruction Loss: 60.8821

Final KL Divergence: 10.9066

Recon/KL Ratio: 5.58

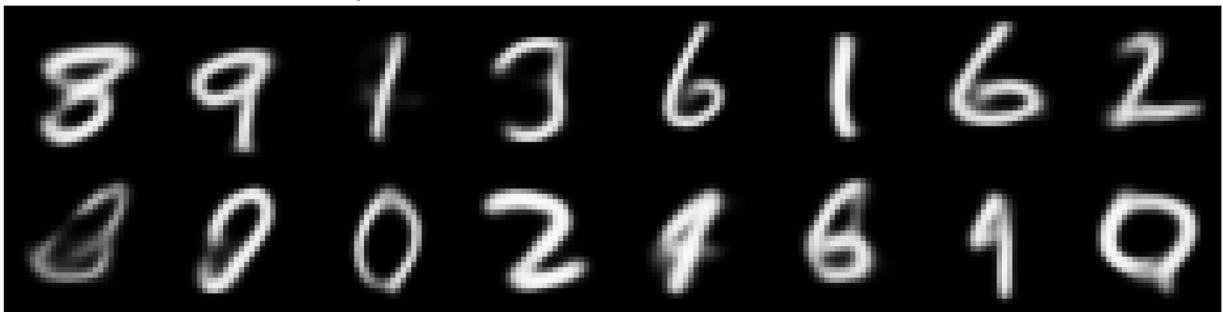
$\beta=5.0$ • Original Samples



$\beta=5.0$ • Reconstructions



$\beta=5.0$ • Random Generations



Experiment summaries:

$\beta=0.5$: Recon=16.5292, KL=41.5967, Recon/KL=0.40

$\beta=1.0$: Recon=24.4601, KL=28.8363, Recon/KL=0.85

$\beta=3.0$: Recon=44.7836, KL=15.1563, Recon/KL=2.95

$\beta=5.0$: Recon=60.8821, KL=10.9066, Recon/KL=5.58

β -VAE Experiment Summary (MNIST)

Qualitative comparison across β settings

β value	Reconstruction loss (relative)	KL divergence (relative)	Reconstruction quality	Generation quality
0.5	lowest	lowest	Sharp digits with high fidelity	Samples resemble training data closely but offer limited variation
1.0	low	moderate	Slightly blurrier yet still faithful digits	Improved diversity with realistic strokes

β value	Reconstruction loss (relative)	KL divergence (relative)	Reconstruction quality	Generation quality
3.0	medium	higher	Noticeable blurring and occasional digit ambiguity	Broader latent usage leading to more varied but less crisp samples
5.0	highest	highest	Reconstructions lose fine details and may misclassify digits	Highly diverse generations, yet realism drops because samples become washed out

The "relative" columns are ranked comparisons: higher β monotonically increases the contribution of the KL term, forcing encoder posteriors to remain close to the unit Gaussian. This inflates the measured KL divergence while preventing the model from allocating capacity to precise reconstructions, so the reconstruction loss grows with β .

Effect on latent representations

Increasing β strengthens the pressure for each latent dimension to match the prior, promoting factorised posteriors. When $\beta \approx 1$ (the conventional VAE), the model balances reconstruction fidelity and latent regularisation, producing latent codes that capture a mixture of shared and disentangled factors. As β is raised (3–5), the KL penalty dominates: encodings become more isotropic and independent, which encourages disentanglement and yields a latent space where traversals manipulate interpretable attributes (stroke thickness, slant, etc.). However, the stronger regularisation also restricts the amount of information each latent dimension can carry, so the decoder receives a less detailed signal and reconstructions appear blurrier.

Conversely, lowering β below 1 relaxes the information bottleneck. The encoder can devote more capacity to per-sample details, restoring sharp reconstructions but collapsing the latent space around training samples, which harms the model's ability to generate diverse digits.

Takeaways

- β provides a knob between reconstruction fidelity and latent factorisation.
- Larger β values promote disentanglement by encouraging isotropic posteriors, but the resulting information bottleneck makes reconstructions less crisp.
- Moderate β (around 1) offers a compromise between faithful reconstructions and useful, moderately disentangled latent dimensions.